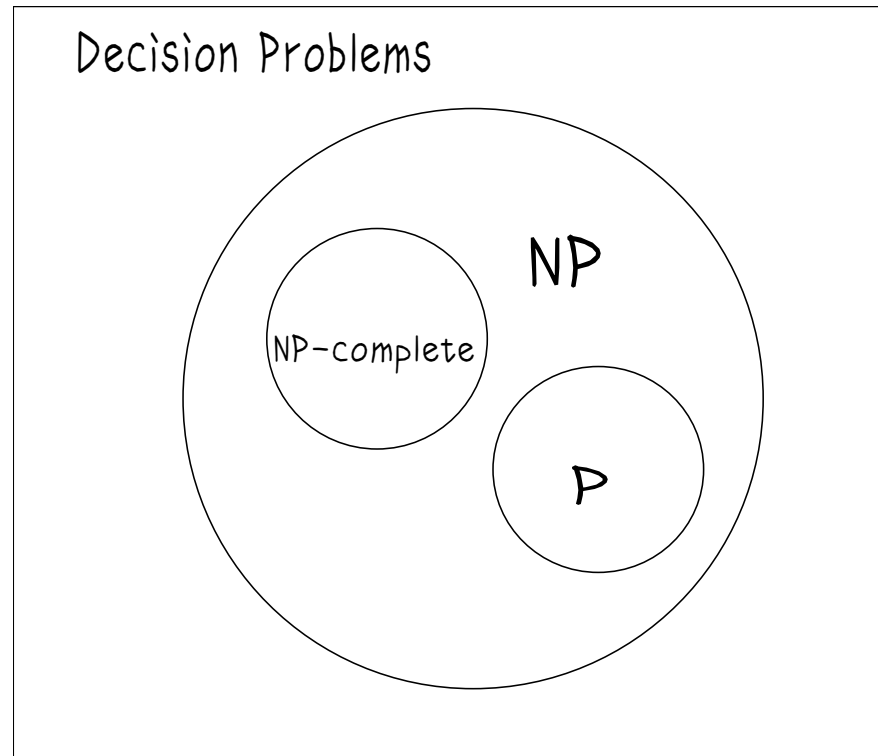


Algorithms and Analysis

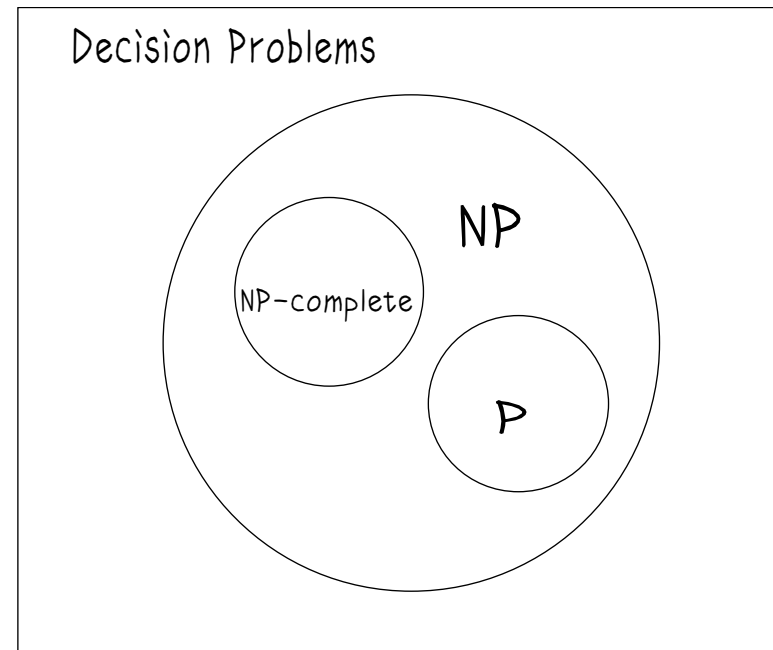
Lesson 23: *Know What's Possible*



Combinatorial optimisation, NP-completeness, polynomial reduction

Outline

1. **Motivation**
2. P, NP and NP-complete
3. Polynomial Reduction



Exponentially Large Search Spaces

- We have seen a large number of decision problems and optimisation problems involving an exponentially large search space
- For some of these we have found efficient algorithms (greedy algorithms, divide and conquer, dynamic programming, . . .)
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Types of Problems

- We concentrate here on two types of problems
 - ★ Decision Problems
 - ★ Combinatorial Optimisation Problems
- Decision problems are problems with a true/false answer, e.g. is it possible to cross all the bridges of Königsberg once?
- We showed earlier that backtracking can be used to find a solution which answers the decision problems, e.g. Hamiltonian circuit problem
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SAT

- Given n Boolean variables $X_i \in \{T, F\}$
- m disjunctive (or's) clauses, e.g.

$$c_1 = X_1 \vee \neg X_2 \vee X_3$$

$$c_2 = \neg X_2 \vee X_3 \vee X_5$$

$$\vdots \quad \quad \vdots$$

$$c_m = X_2 \vee \neg X_4 \vee \neg X_5$$

- Find an assignment, $\mathbf{X} \in \{T, F\}^n$ which satisfies all the clauses
- We can view this as finding an assignment that makes the formula $f(\mathbf{X})$ true where

$$f(\mathbf{X}) = c_1 \wedge c_2 \wedge \cdots \wedge c_m$$

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- Often we can cast a decision problem as an optimisation problem
- E.g. the MAX-SAT problem is to find an assignment of variables that satisfies the most clauses
- If we can solve the MAX-SAT optimisation problem we can solve the decision problem
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Combinatorial Optimisation Problems

- In the set of discrete optimisation problems an important class are those that involve *combinatorial objects* such as permutations, binary string, etc.
- Optimisation problems involving such objects are termed **combinatorial optimisation problems**
- Classical examples of such problems include
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 - ★ Graph colouring
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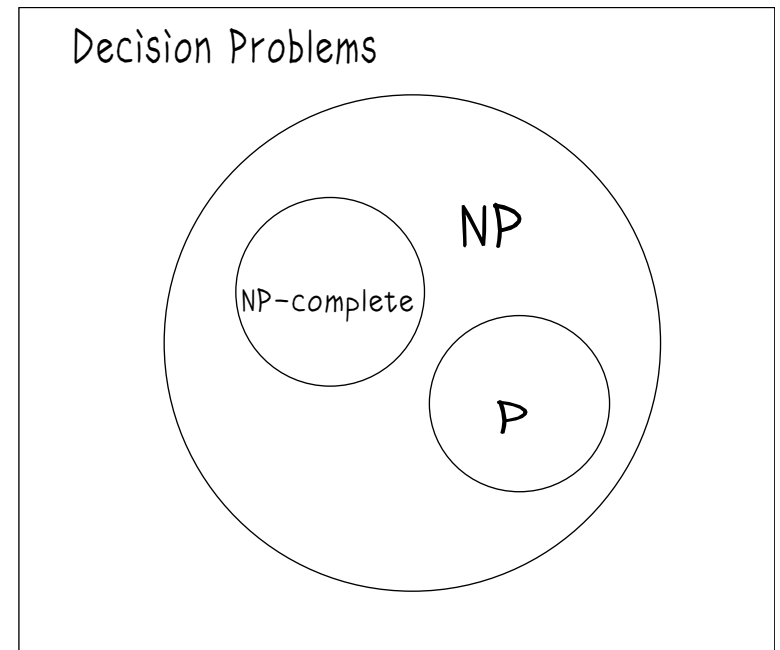
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- Answer:
- However, no one has discovered such an algorithm and if they do it will have huge implications
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Decision Problems

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- E.g. does a TSP have a tour less than 2 000 miles?
- Algorithmic complexity theory deals with classes of decision problems that have some characteristic size, n
- E.g. n is the number of cities in a TSP
- Any decision problem that can be answered with an algorithm that runs in polynomial time (n^a) on a normal computer (Turing machine) is said to be in class P
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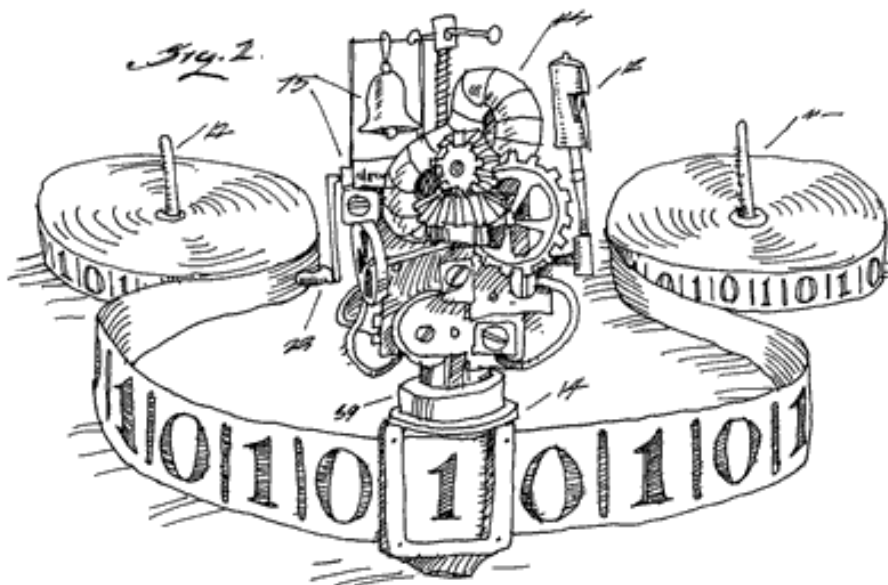
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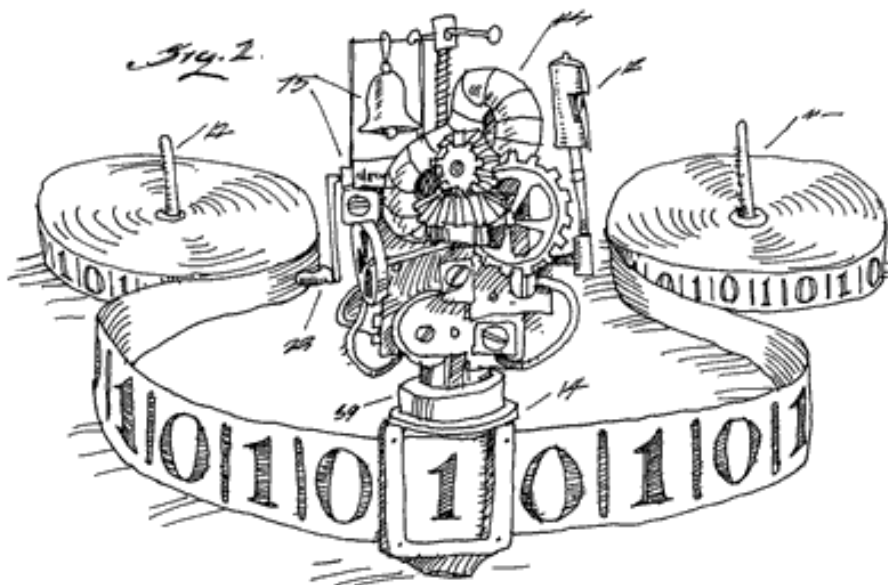
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- All decision problems with polynomial algorithms are also in class NP ($P \subset NP$)
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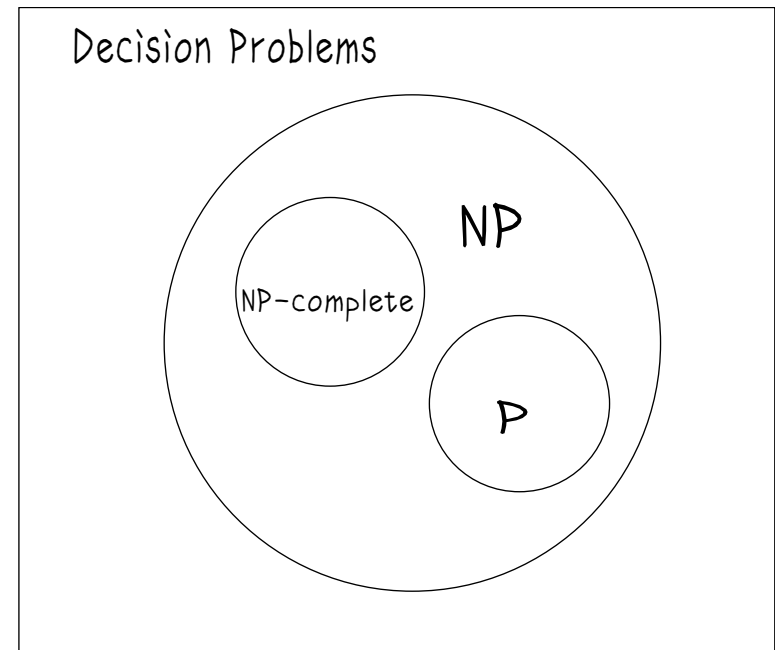
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- If the clauses are simultaneously satisfiable then there exists an input that satisfies the conditions

Idea Behind Cook's Theorem

- Cook showed that a non-deterministic Turing machine could be encoded as a big SAT formula
- The evolution of the state and tape was represented by a big tableau ($n^k \times n^k$ -table where n^k is the time it takes for the Turing machine to verify the answer)
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Outline

1. Motivation
2. P, NP and NP-complete
3. **Polynomial Reduction**



Polynomial Reductions

- Given two decision problems A and B we say there is a **polynomial reduction** from A to B if
 - ★ Every instance of A can be mapped to an instance of B:
 - ★ The truth of the instance A is the same as the corresponding instance B
- We can therefore use B to solve A
- So: $B \in P \rightarrow A \in P$
- The contrapositive of this statement is
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SAT to 3-SAT

- We can reduce a clause with 4 variable to a clause with 3

$$X_1 \vee \neg X_3 \vee X_6 \vee \neg X_{10} \equiv (X_1 \vee \neg X_3 \vee Z) \wedge (\neg Z \vee X_6 \vee \neg X_{10})$$

- In doing so we increase the number of variables and the number of clauses to satisfy
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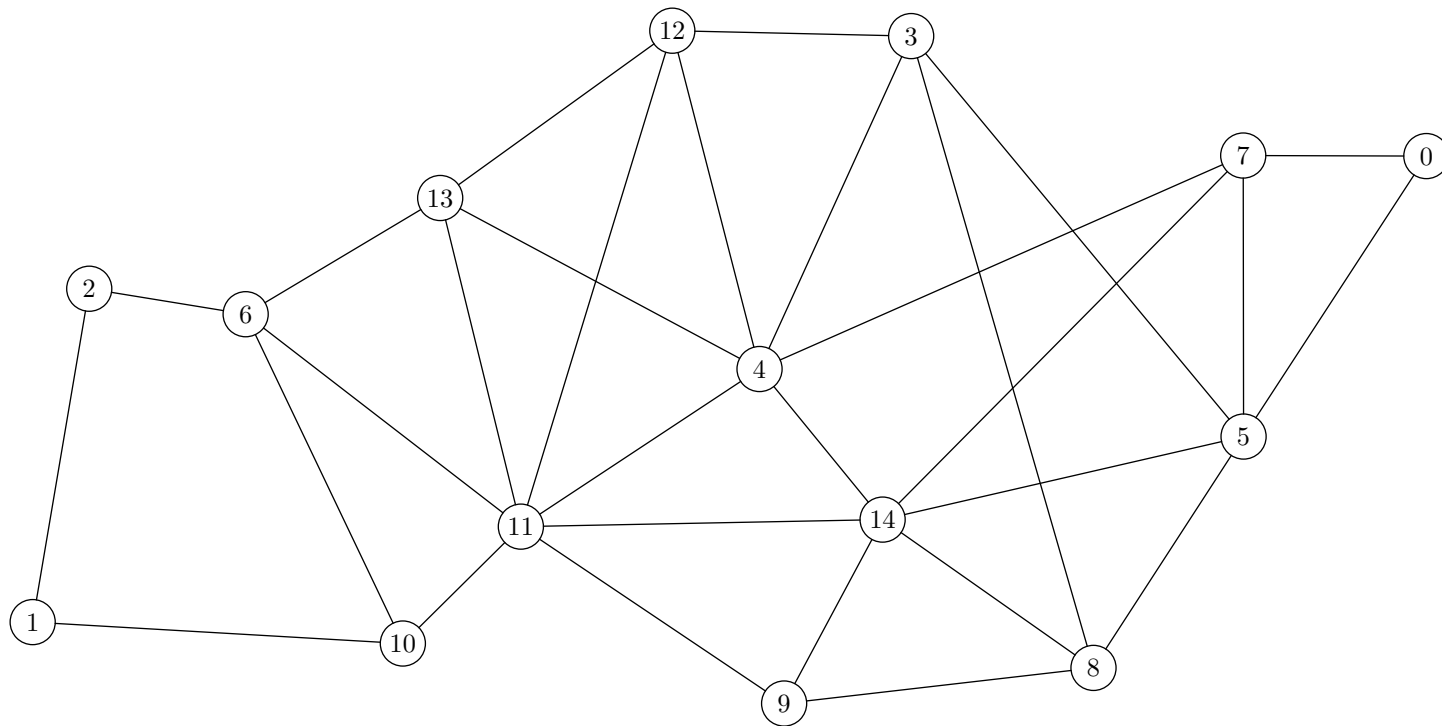
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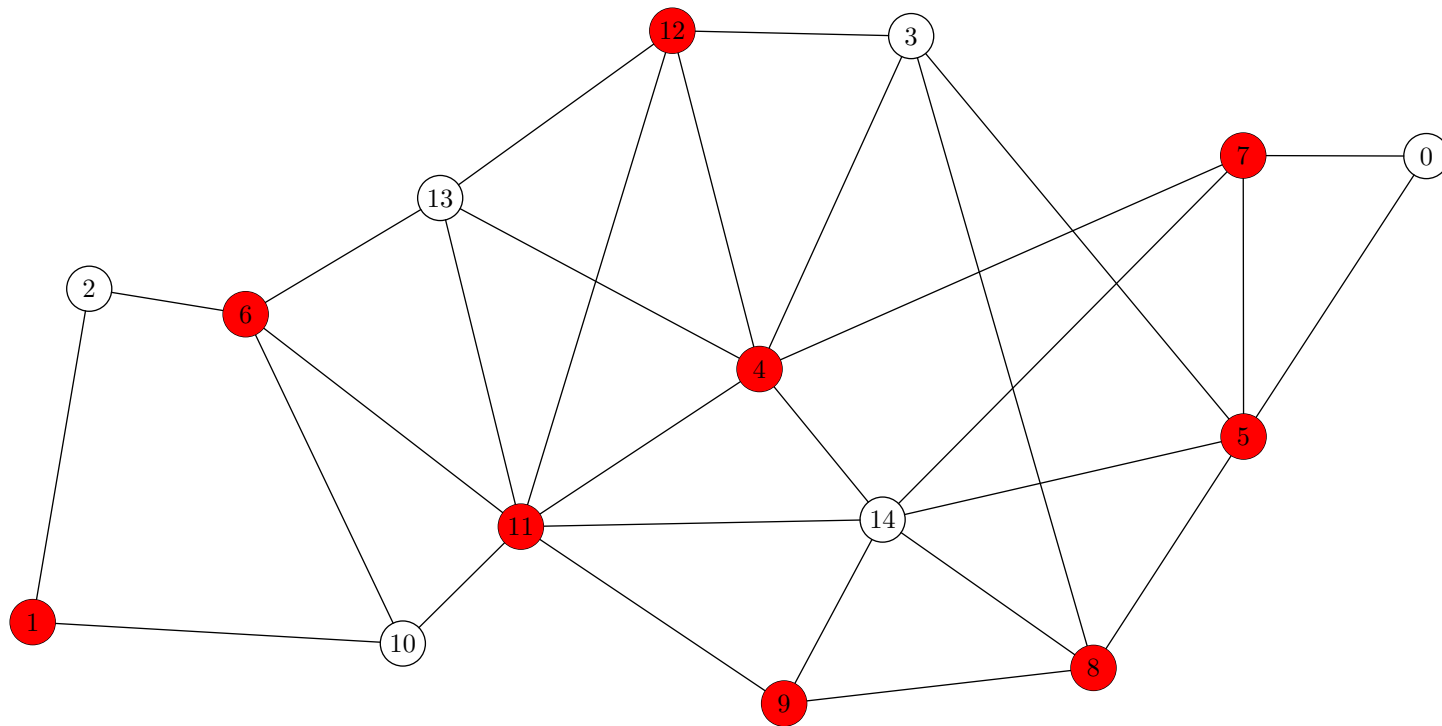
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- To show vertex cover is NP-complete we show that every instance of 3-SAT is reducible to vertex cover
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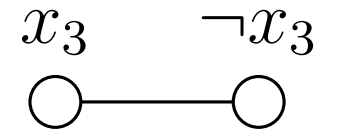
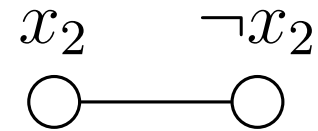
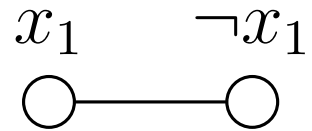
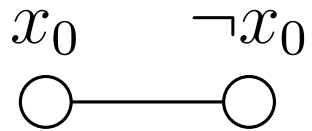
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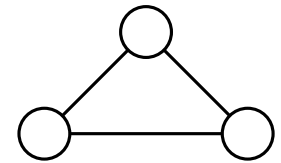
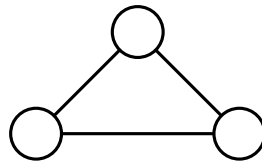
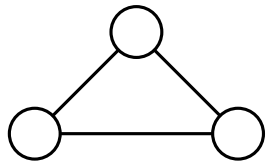
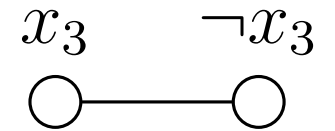
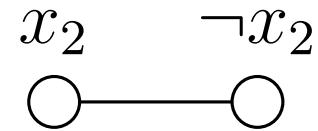
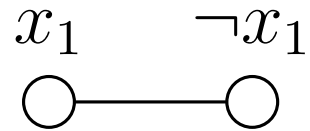
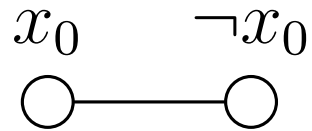
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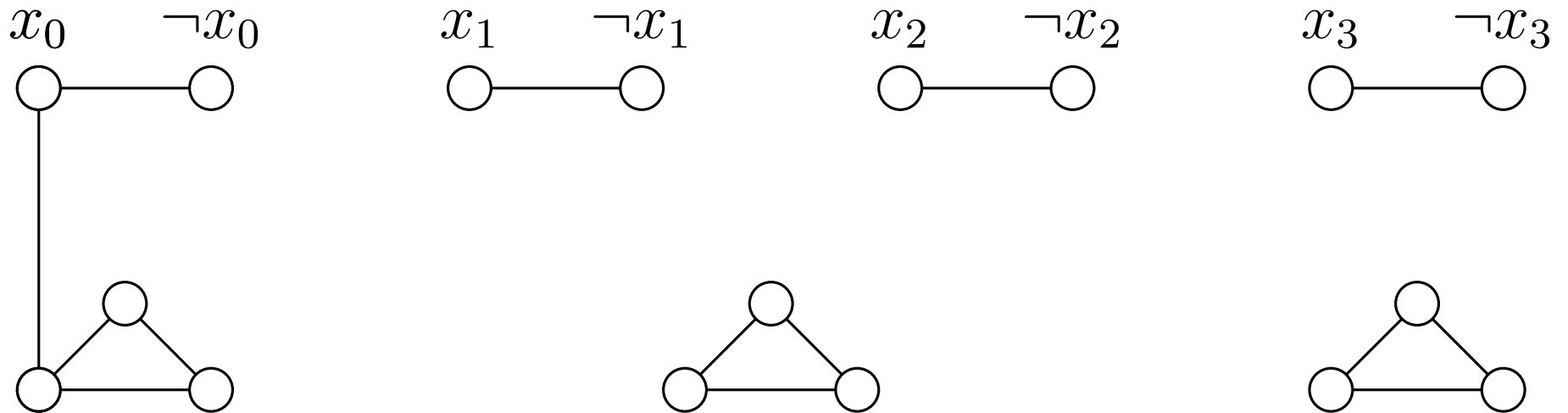
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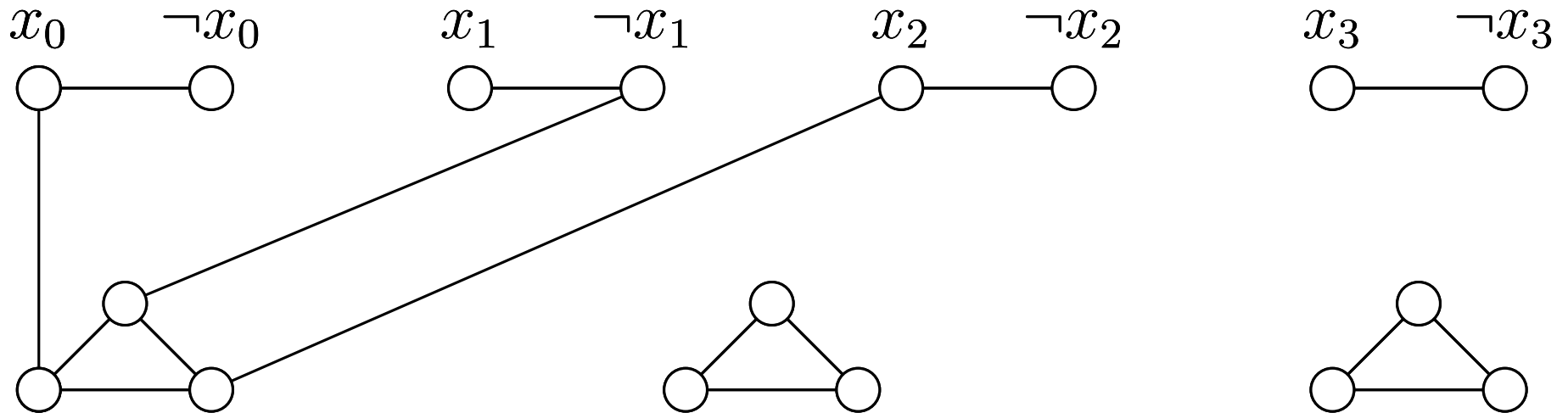
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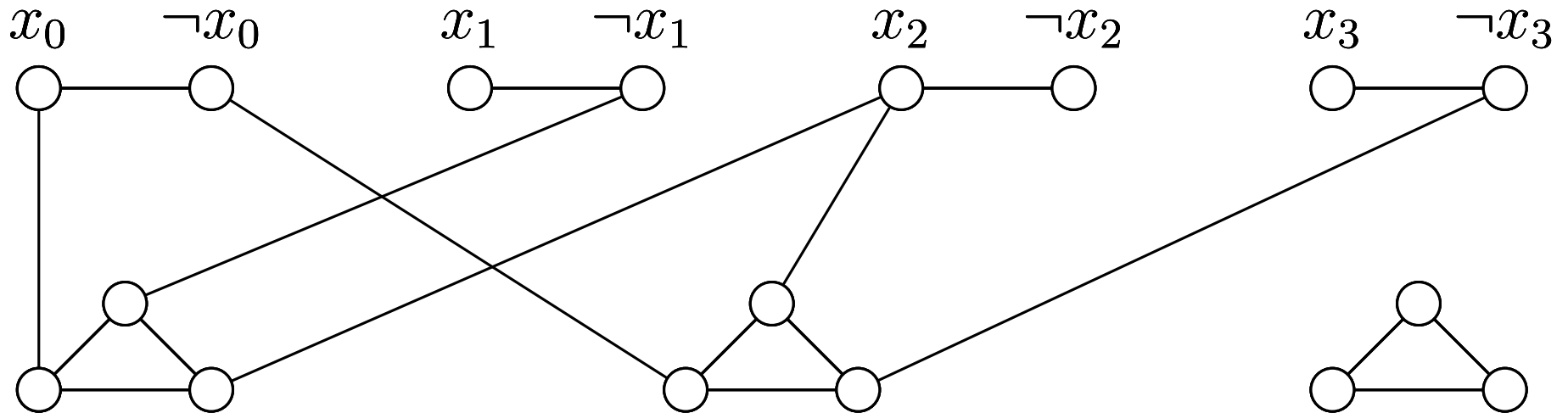
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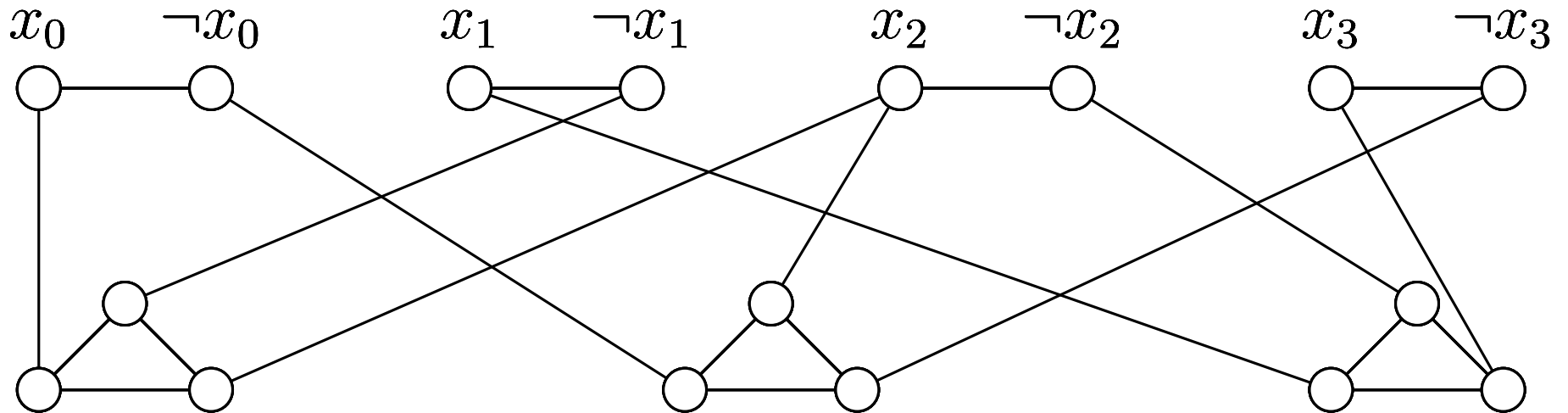
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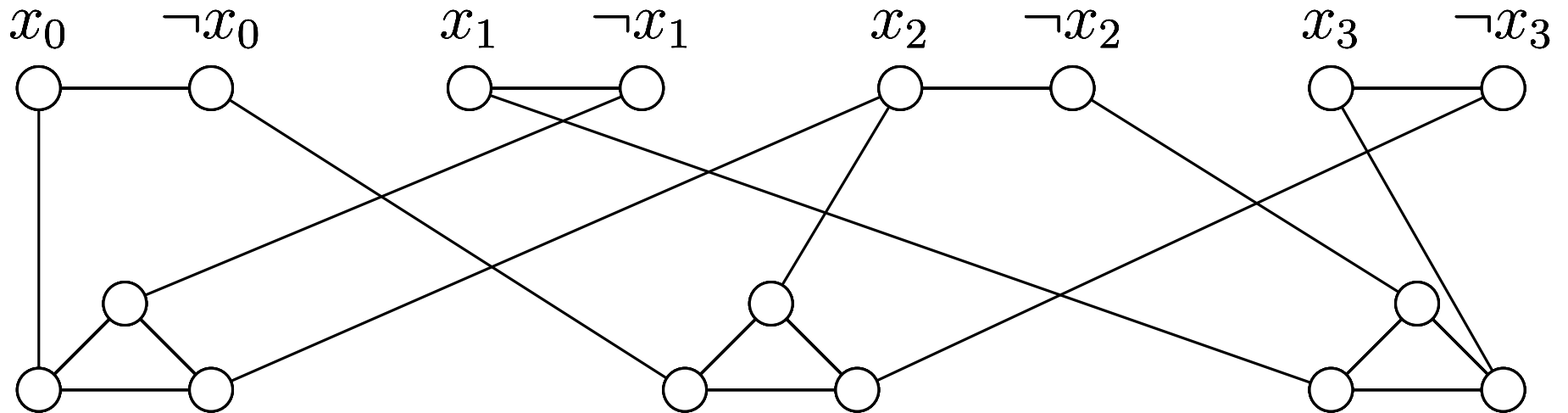
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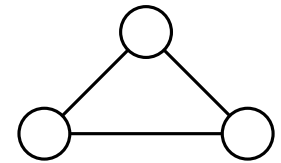
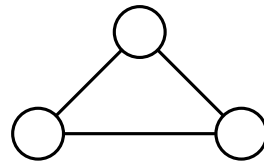
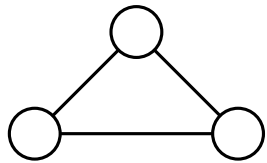
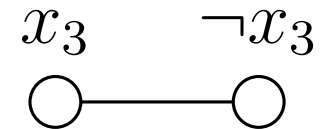
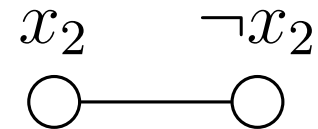
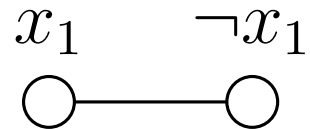
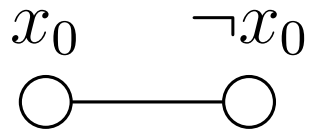
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Cover all edges using $n + 2m = 4 + 2 \times 3 = 10$ vertices

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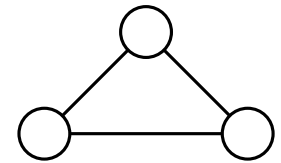
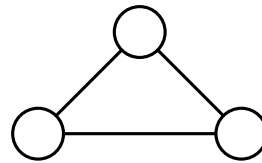
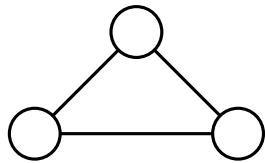
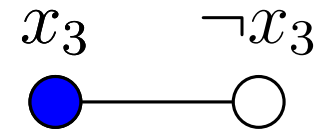
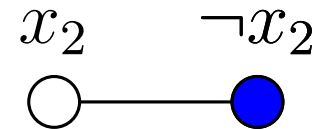
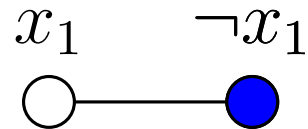
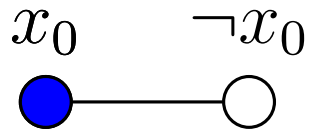
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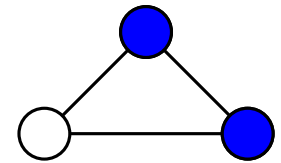
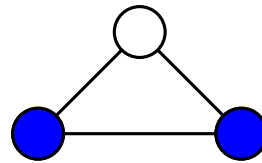
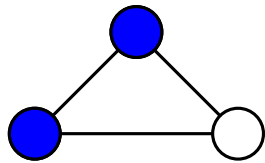
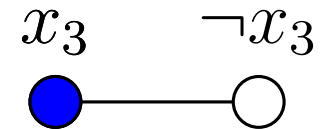
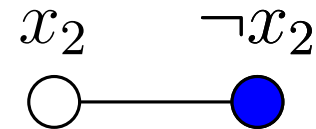
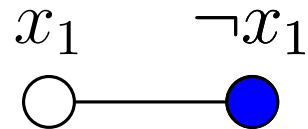
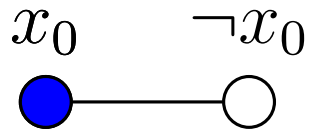
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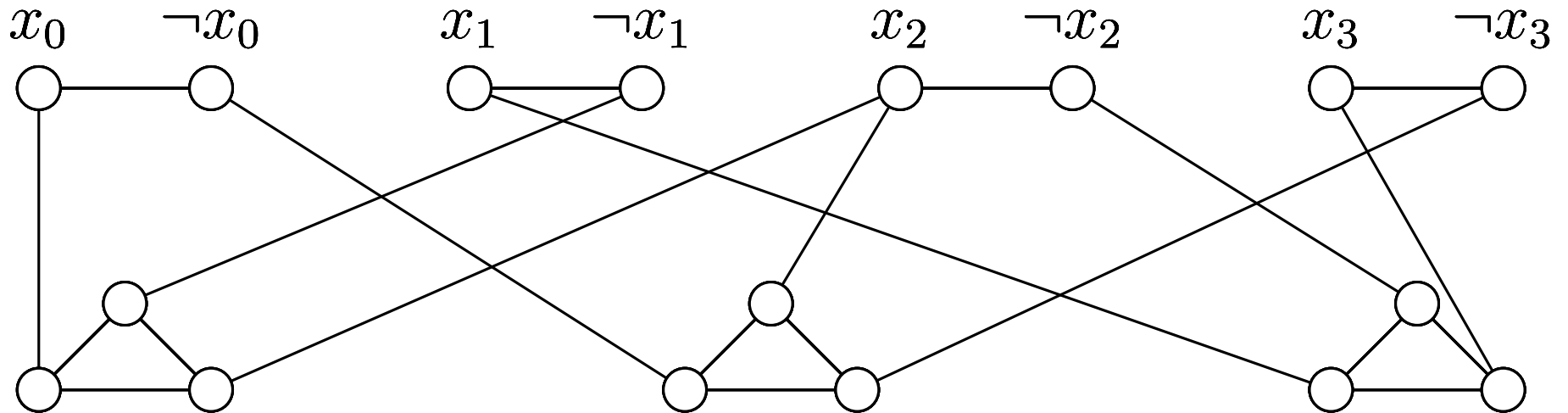
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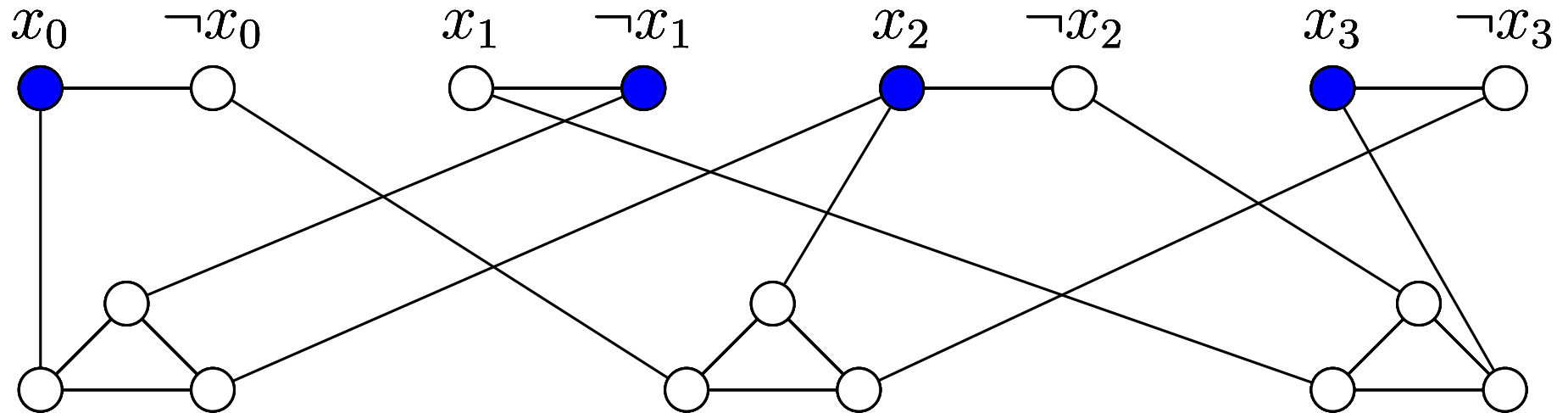
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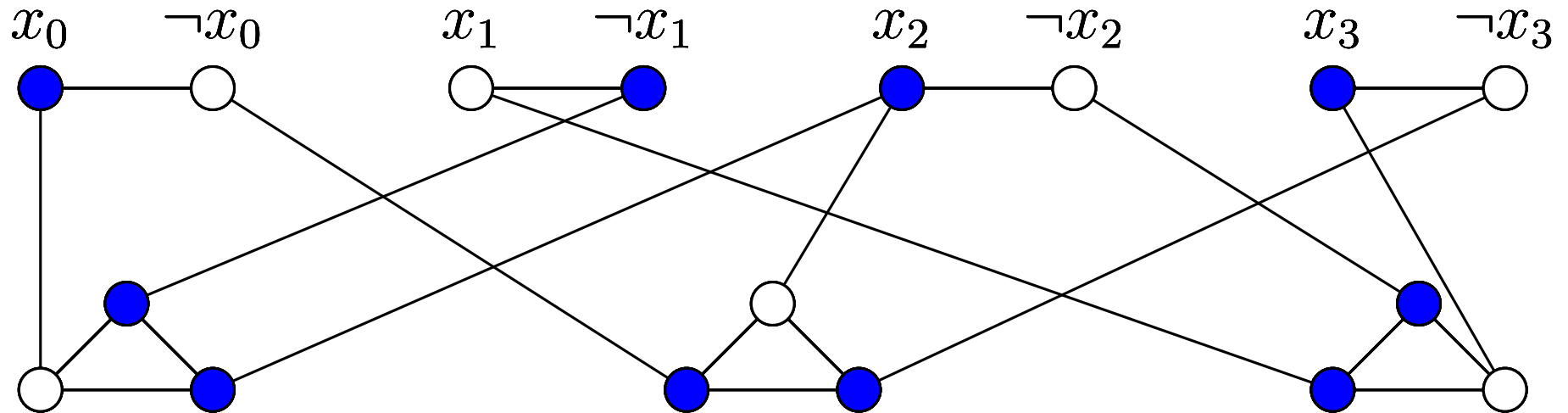
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- As we can polynomial reduce any instance of SAT to vertex cover then vertex cover is also NP-complete
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 - ★ Many scheduling problems
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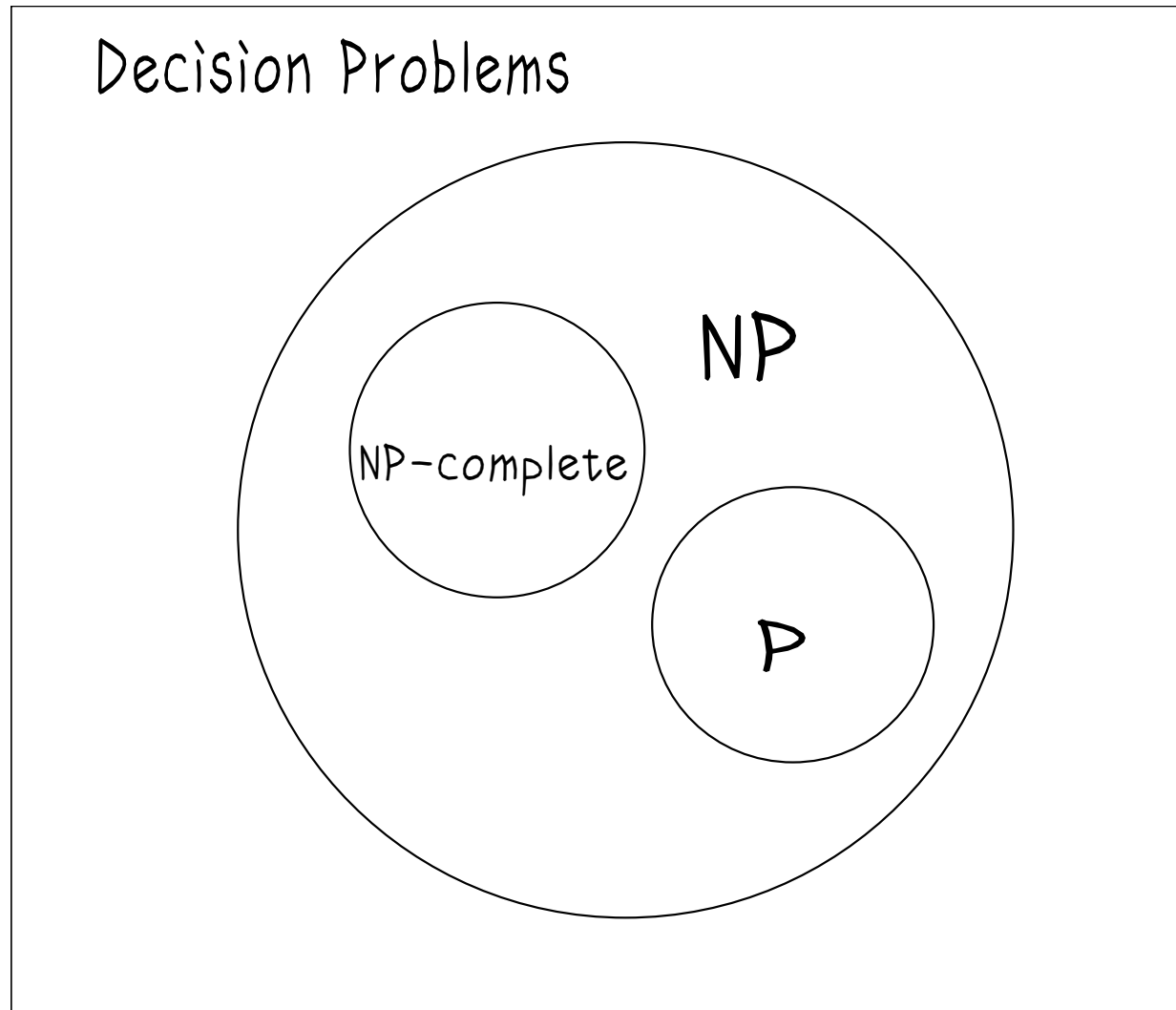
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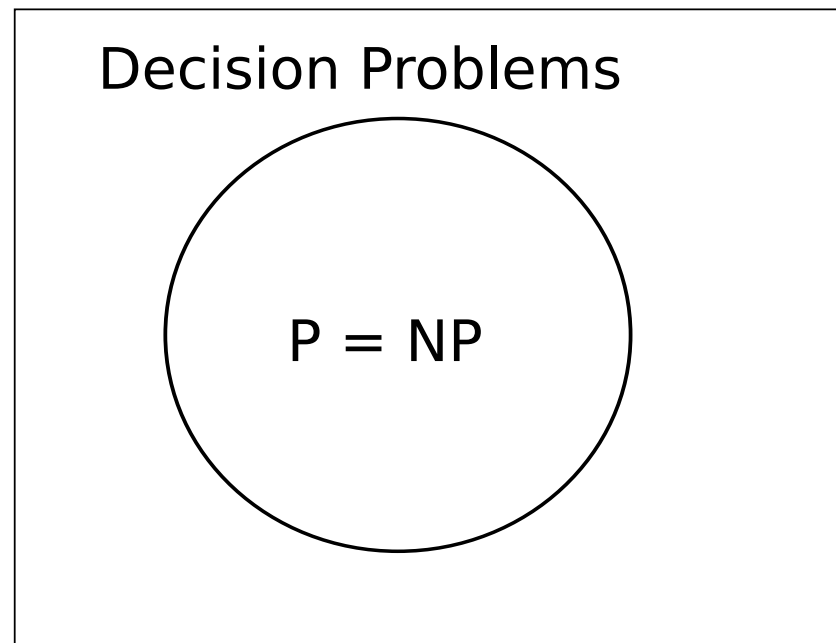
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 - ★ Quadratic integer problems

Structure of Decision Problems



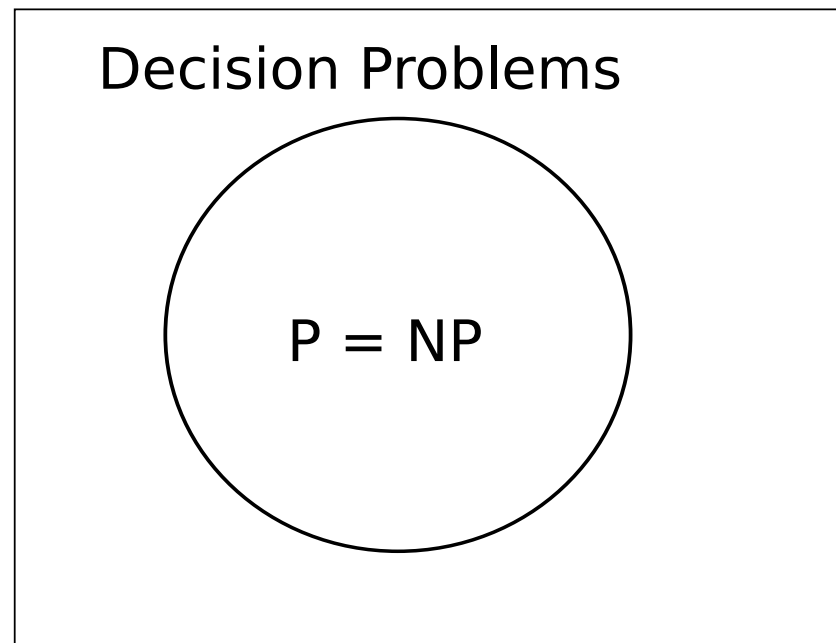
$$P \neq NP?$$

- No one has proved that any problem in NP is not solvable in polynomial time
- If any NP-complete problem was solved in polynomial time all problems in NP would be solve and $NP=P$



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NP-Hard

- TSP is not a decision problem—although we can make it into one—Is there a tour shorter than L ?
- However, if we can find the shortest tour in polynomial time we could solve the TSP decision problem
- Thus finding the shortest tour is at least as hard as solving the decision problems
- Problems that are at least as hard as NP-complete decision problems are said to be in **NP-hard**
- Graph colouring (finding a colouring with the least number of conflicts), job scheduling, etc. are all examples of NP-hard problems

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- Graph isomorphism, GI, (are two graphs identical up to a relabelling of the vertices?) has not been proved to be NP-complete—it is postulated that

$$GI \in NP \wedge GI \notin P \wedge GI \notin \text{NP-complete}$$

- Factoring is *not* believed to be NP-hard, but it is believed to be sufficiently hard that most banks use an encryption technique based on people not being able to factor large numbers easily
- For large problems polynomial algorithms can take too long

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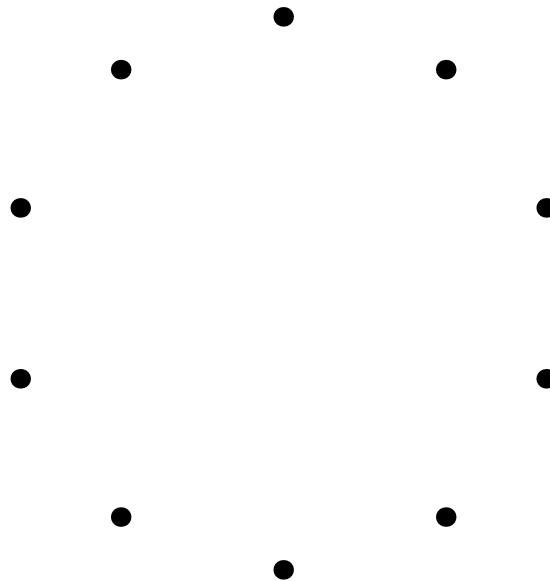
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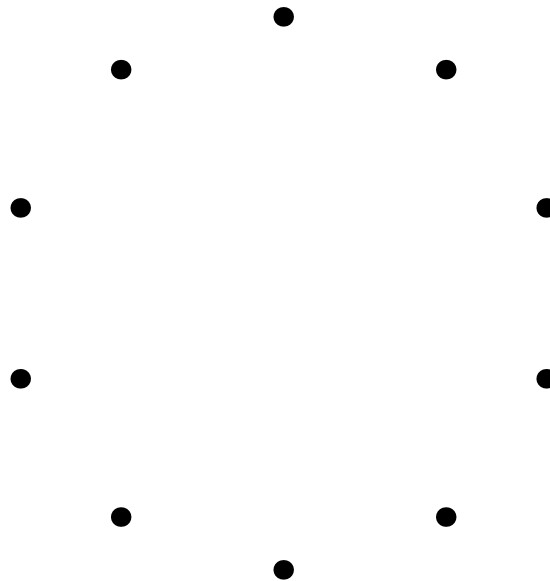
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- NP-hardness is a worst case analysis
- It means there exist some instance of the problem that we don't know how to solve in polynomial time
- Many instance of the problem might be rather easy to solve
- What is the optimal TSP tour for the problem below?



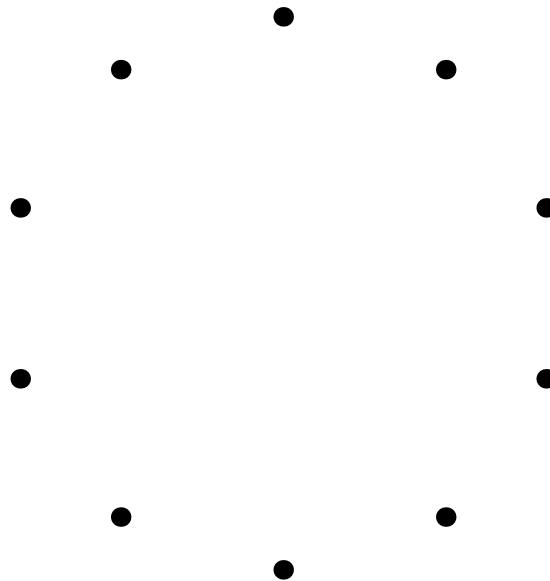
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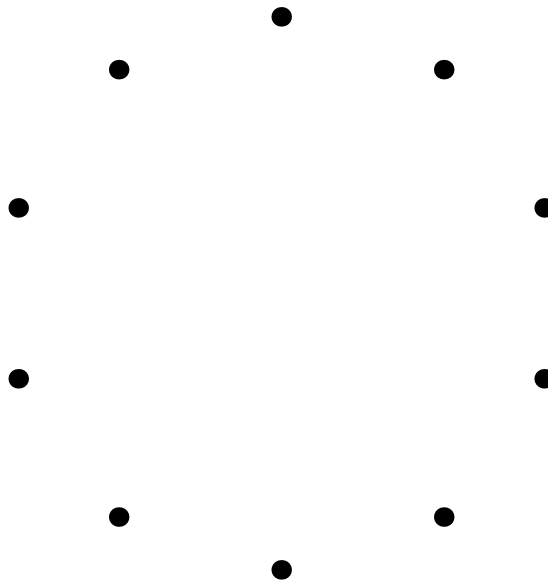
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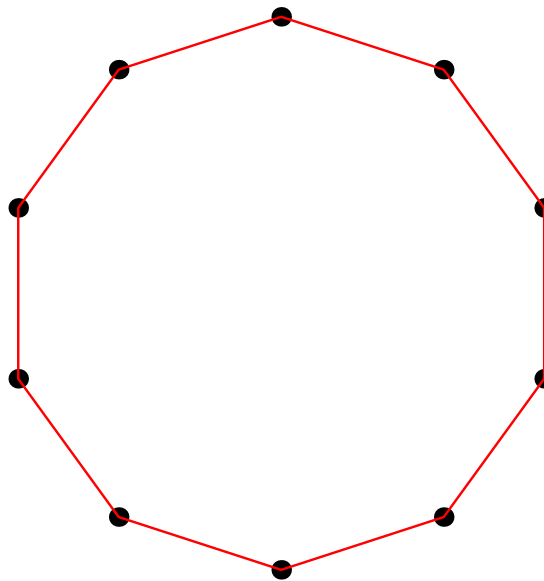
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 - ★ Given a set of numbers find a subset whose sums is as close as possible to some constant
 - ★ Subset-sum is in NP-hard but there exist a “pseudo-polynomial time” algorithm which solves almost every instance in polynomial time
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- . . . but probably not for all
- There are no known polynomial algorithm for any NP-complete problem
- These include many famous problems: TSP, graph-colouring, scheduling, . . .
- If you could find a polynomial algorithm for any of these problems then you could use it to solve all problems in NP in polynomial time

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